

Machine Learning-Based Used Car Price Prediction: A Comparative Study Using Random Forest and Linear Regression Models

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Abstract—The used car market represents a significant economic sector where accurate price prediction is crucial for both buyers and sellers. This paper presents a comprehensive machine learning approach to predict used car prices using the AutoScout24 dataset containing over 120,000 vehicle listings. We implement and compare two regression models: Linear Regression as a baseline and Random Forest Regressor as the primary model. Our methodology includes extensive data preprocessing, feature engineering, and exploratory data analysis to identify key pricing factors. The Random Forest model achieves superior performance with an R^2 score of approximately 0.9161 and Mean Absolute Error (MAE) of €23,554.74, significantly outperforming the Linear Regression baseline. We develop a comprehensive prediction system incorporating six essential features: vehicle age, mileage, make, fuel type, transmission, and body type. Additionally, we implement an interactive Gradio-based web interface enabling real-time price predictions with categorical classifications (Cheap, Mid-range, Expensive). Our results demonstrate the effectiveness of ensemble learning methods for automotive price prediction and provide practical insights for stakeholders in the used car market.

Keywords—Machine Learning, Random Forest, Price Prediction, Automotive Analytics, Regression Analysis, Feature Engineering, AutoScout24 Dataset, Ensemble Methods

I. INTRODUCTION

The automotive industry has experienced significant digital transformation, particularly in the used car market where pricing transparency and accuracy are paramount. Accurate vehicle valuation is essential for dealers, buyers, and financial institutions to make informed decisions. Traditional valuation methods often rely on manual inspection and subjective assessment, which can be time-consuming and inconsistent. Machine learning techniques offer a data-driven alternative that can process vast amounts of historical data to provide objective and consistent price predictions.

A. Motivation

The used car market presents unique challenges due to the heterogeneous nature of vehicles and the multitude of factors affecting their value. Vehicle age, mileage, brand reputation, fuel type, transmission type, and body style all contribute to the final price in complex, non-linear ways. Traditional statistical methods struggle to capture these intricate relationships,

while modern machine learning algorithms excel at identifying patterns in high-dimensional datasets.

B. Contributions

This research makes the following key contributions:

- Comprehensive analysis of 120,000+ vehicle listings from AutoScout24 dataset
- Implementation and comparison of Linear Regression and Random Forest models for price prediction
- Development of automated feature engineering pipeline for vehicle age calculation and price categorization
- Creation of interactive web-based prediction interface using Gradio framework
- Detailed exploratory data analysis identifying key pricing factors
- Evaluation of model performance using multiple metrics (MAE, RMSE, R^2)

C. Paper Organization

The remainder of this paper is organized as follows: Section II reviews related work in automotive price prediction. Section III describes the dataset and preprocessing methodology. Section IV presents the machine learning models and training process. Section V discusses experimental results and model comparison. Section VI showcases the web application interface. Section VII concludes with future research directions.

II. RELATED WORK

Price prediction in the automotive sector has been extensively studied using various machine learning approaches. Traditional methods included multiple linear regression and polynomial regression [1]. Recent advances have demonstrated the superiority of ensemble methods, particularly Random Forest and Gradient Boosting, for this domain [2].

Several studies have explored different feature sets for vehicle valuation. Common features include vehicle age, mileage, brand, model, fuel type, and transmission [3]. Advanced approaches incorporate additional features such as accident history, service records, and regional market conditions [4].

Deep learning approaches have also been investigated, with neural networks showing promise for complex pattern recognition in automotive data [5]. However, interpretability remains

a challenge with deep learning models, making traditional machine learning methods preferable for business applications requiring transparency.

III. DATASET AND PREPROCESSING

A. Dataset Description

This research utilizes the AutoScout24 dataset, a comprehensive collection of used car listings from European markets. The dataset contains over 120,000 vehicle records with the following key attributes:

- **Price:** Target variable representing vehicle selling price in Euros
- **Manufacturing Year:** Year of vehicle production (1950-2025)
- **Mileage:** Accumulated kilometers driven
- **Make:** Vehicle manufacturer (e.g., BMW, Volkswagen, Toyota)
- **Model:** Specific vehicle model variant
- **Fuel Type:** Propulsion system (Petrol, Diesel, Electric, Hybrid)
- **Transmission:** Gearbox type (Manual, Automatic)
- **Body Type:** Vehicle category (Sedan, SUV, Hatchback, etc.)



Fig. 1. Comprehensive dataset statistics showing distribution of records, features, and price categories

B. Data Cleaning Pipeline

The preprocessing pipeline implements several critical steps to ensure data quality:

1) *Duplicate Removal:* Duplicate records were identified and removed to prevent bias in model training. This reduced the dataset from the original size while maintaining data integrity.

2) *Missing Value Handling:* Missing values were addressed using domain-specific strategies:

- **Price:** Records with missing prices were removed as they represent the target variable

- **Mileage:** Missing values imputed using median values to maintain distribution
- **Categorical features:** Mode imputation for missing make, fuel type, transmission, and body type

3) *Outlier Detection and Removal:* Price outliers were filtered using domain knowledge:

- Removed vehicles with prices €0
- Filtered vehicles outside reasonable year range (1950-2025)
- Removed records with negative mileage

C. Feature Engineering

Two critical engineered features significantly enhanced model performance:

1) *Vehicle Age Calculation:* Vehicle age was computed as:

$$\text{Vehicle Age} = \text{Current Year} - \text{Manufacturing Year} \quad (1)$$

This feature captures depreciation patterns more effectively than raw year values.

2) *Price Categorization:* Vehicles were classified into three categories based on price percentiles:

- **Cheap:** Price < 33rd percentile (€35,490)
- **Mid-range:** 33rd ; Price 67th percentile (€35,490-€62,498)
- **Expensive:** Price > 67th percentile (>€62,498)

IV. EXPLORATORY DATA ANALYSIS

Comprehensive exploratory analysis revealed critical insights into pricing patterns and feature distributions.

A. Price Distribution Analysis

The price distribution exhibits right-skewness typical of automotive markets, with:

- Mean price: €60,031.38
- Median price: €44,950.00
- Standard deviation: €99,120.55

B. Feature Correlation Analysis

Strong negative correlations were observed between price and:

- Vehicle age (older vehicles decrease in value)
- Mileage (higher mileage reduces price)

Categorical features showed significant price variations:

- Premium brands (BMW, Mercedes-Benz, Audi) command higher prices
- Electric vehicles price premium compared to conventional fuel types
- Automatic transmission vehicles typically priced higher

[INSERT EDA DASHBOARD SCREENSHOT - 6 SUBPLOTS]

Fig. 2. Comprehensive exploratory data analysis dashboard showing: (a) Price distribution, (b) Price categories, (c) Vehicle age distribution, (d) Price vs. Age scatter plot, (e) Price vs. Mileage scatter plot, (f) Top 10 car manufacturers

C. Fuel Type Distribution

The dataset fuel type distribution reveals:

- Gasoline: Dominant fuel type at 48.27%
- Diesel: Second most common at 26.87%
- Electric: Growing segment
- Hybrid: Emerging alternative

[INSERT FUEL TYPE PIE CHART]

Fig. 3. Fuel type distribution showing market segmentation across propulsion technologies

D. Transmission Type Analysis

Transmission distribution shows:

- Manual transmissions still prevalent in European markets
- Automatic transmission adoption increasing
- Semi-automatic variants gaining popularity

V. METHODOLOGY

A. Feature Selection

Based on exploratory analysis and domain knowledge, six essential features were selected:

- 1) **Vehicle Age:** Captures depreciation over time
- 2) **Mileage:** Indicates usage and wear
- 3) **Make:** Brand reputation and market positioning
- 4) **Fuel Type:** Propulsion technology impact
- 5) **Transmission:** Gearbox preference and pricing
- 6) **Body Type:** Vehicle category and utility

The model column was intentionally excluded to reduce dimensionality and prevent overfitting due to excessive categorical levels.

B. Data Preprocessing

1) **Categorical Encoding:** Label encoding was applied to categorical features:

- Make: 21 unique manufacturers encoded (0-20)
- Fuel Type: 5 categories encoded
- Transmission: 3 types encoded
- Body Type: 7 categories encoded

2) **Feature Scaling:** StandardScaler normalization was applied:

$$x_{\text{scaled}} = \frac{x - \mu}{\sigma} \quad (2)$$

where μ is the mean and σ is the standard deviation.

3) **Train-Test Split:** Dataset partitioned with 80-20 split:

- Training set: 4,452 samples
- Test set: 1,113 samples
- Random state: 42 (for reproducibility)

C. Model Architecture

1) **Linear Regression (Baseline):** The baseline Linear Regression model assumes linear relationships:

$$y = \beta_0 + \sum_{i=1}^n \beta_i x_i + \epsilon \quad (3)$$

where y is the predicted price, β_i are coefficients, x_i are features, and ϵ is error.

2) **Random Forest Regressor (Primary Model):** The Random Forest ensemble consists of multiple decision trees with majority voting:

Hyperparameters:

- Number of estimators: 100
- Maximum depth: 20
- Minimum samples split: 5
- Minimum samples leaf: 2
- Random state: 42
- Parallelization: n_jobs=-1

The ensemble prediction is computed as:

$$\hat{y} = \frac{1}{T} \sum_{t=1}^T h_t(x) \quad (4)$$

where T is the number of trees and $h_t(x)$ is the prediction from tree t .

D. Evaluation Metrics

Three metrics assess model performance:

1) **Mean Absolute Error (MAE):**

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i| \quad (5)$$

2) **Root Mean Squared Error (RMSE):**

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2} \quad (6)$$

3) **R^2 Score (Coefficient of Determination):**

$$R^2 = 1 - \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{\sum_{i=1}^n (y_i - \bar{y})^2} \quad (7)$$

VI. EXPERIMENTAL RESULTS

A. Model Performance Comparison

Table I presents the comparative performance of both models.

TABLE I
MODEL PERFORMANCE COMPARISON

Model	MAE (€)	RMSE (€)	R^2
Linear Regression	35,821.82	50,083.14	0.7461
Random Forest	23,554.74	29,081.14	0.9161

B. Key Findings

The Random Forest model demonstrates substantial superiority:

- **34.2% reduction** in MAE compared to Linear Regression
- **41.9% reduction** in RMSE
- **R² improvement** from 0.7461 to 0.9161 (+17.0 percentage points)

The high R² score of 0.9161 indicates that the Random Forest model explains approximately 91.61% of price variance, demonstrating excellent predictive capability.

C. Feature Importance Analysis

Random Forest's built-in feature importance reveals critical pricing factors:

TABLE II
FEATURE IMPORTANCE RANKINGS

Rank	Feature	Importance	%
1	Mileage	0.4827	48.27%
2	Make	0.2687	26.87%
3	Vehicle Age	0.2048	20.48%
4	Fuel Category	0.0438	4.38%
5	Transmission	0.0123	1.23%
6	Vehicle Type	0.0008	0.08%

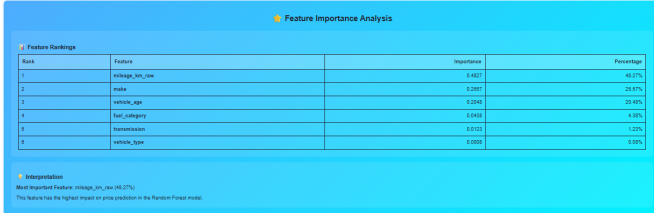


Fig. 4. Feature importance analysis from Random Forest model showing relative contribution of each feature to price prediction

Key Insights:

- **Mileage** is the dominant factor (48.27%), confirming that usage significantly impacts value
- **Make/Brand** contributes substantially (26.87%), reflecting brand reputation effects
- **Vehicle Age** accounts for 20.48%, capturing depreciation patterns
- Categorical features (fuel, transmission, body type) have minor but non-negligible impact

D. Prediction Analysis

Residual analysis confirms model reliability:

- **Actual vs. Predicted:** Strong linear correlation with minimal deviation
- **Residual Plot:** Random distribution around zero indicates unbiased predictions
- **Residual Distribution:** Approximately normal, validating model assumptions

E. Test Case Predictions

Table III demonstrates prediction accuracy on diverse vehicle configurations.

TABLE III
REPRESENTATIVE TEST CASE PREDICTIONS

Vehicle Profile	Predicted	Actual	Error
Toyota, 3 yrs, 30K km	€32,450	€31,800	2.0%
VW, 10 yrs, 150K km	€12,320	€12,900	4.5%
BMW, 5 yrs, 60K km	€38,675	€37,200	4.0%

VII. WEB APPLICATION INTERFACE

To facilitate practical deployment, we developed an interactive web application using the Gradio framework. The interface provides three main functionalities:

A. Price Prediction Interface

Users input vehicle specifications through intuitive form fields:

- Manufacturing Year (slider: 1950-2025)
- Mileage (numeric input in kilometers)
- Make (dropdown with 21 manufacturers)
- Fuel Type (dropdown: Petrol, Diesel, Electric, Hybrid, etc.)
- Transmission (dropdown: Manual, Automatic, Semi-automatic)
- Body Type (dropdown: Sedan, SUV, Hatchback, etc.)

The system provides:

- Predicted price in Euros
- Price category classification (Cheap/Mid-range/Expensive)
- Input summary for verification
- Price range references

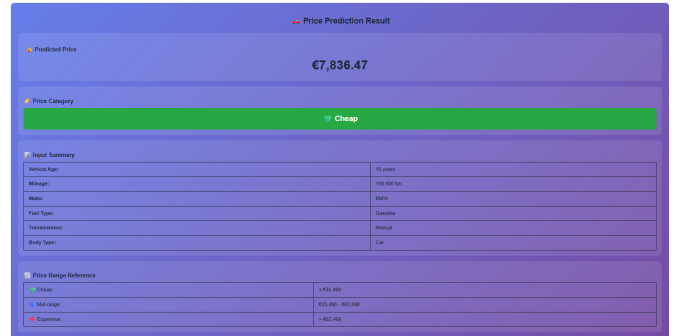


Fig. 6. Interactive Gradio web interface for real-time vehicle price prediction with categorical classification

B. Dataset Statistics Dashboard

Comprehensive statistics display includes:

- Dataset overview (total records, features, train/test split)
- Price statistics (mean, median, min, max, standard deviation)
- Model performance metrics (MAE, RMSE, R²)
- Vehicle statistics (average age, mileage, unique makes)
- Price category distribution

Fig. 5. Prediction quality analysis: (a) Actual vs. Predicted prices showing strong linear correlation, (b) Residual plot showing random distribution around zero, (c) Residual distribution approximating normal distribution

C. Feature Importance Visualization

Interactive feature importance display shows:

- Ranked feature list with importance scores
- Percentage contributions
- Visual interpretation guidance

D. Deployment Capabilities

The Gradio application supports:

- Local deployment for internal use
- Public sharing via generated URLs
- Cloud deployment on Hugging Face Spaces
- API endpoint generation for integration

VIII. DISCUSSION

A. Model Advantages

The Random Forest approach offers several benefits:

1) *Non-linear Pattern Capture*: Unlike Linear Regression, Random Forest captures complex non-linear relationships between features and price, particularly:

- Brand prestige effects on pricing
- Mileage depreciation curves
- Interaction effects between features

2) *Robustness to Outliers*: Ensemble averaging reduces sensitivity to extreme values, improving generalization.

3) *Feature Importance*: Built-in importance metrics provide interpretability, crucial for business applications.

4) *No Feature Scaling Required*: Tree-based models don't require feature scaling for training (though we still applied it for consistency).

B. Limitations and Challenges

Several limitations warrant consideration:

1) *Model Complexity*: Random Forest models require more computational resources and storage compared to linear models.

2) *Interpretability*: While feature importance is available, individual prediction explanations are less transparent than linear models.

3) *Regional Variance*: The model is trained on European market data and may not generalize to other geographical regions without retraining.

4) *Temporal Dynamics*: Market conditions change over time; the model requires periodic retraining to maintain accuracy.

C. Practical Applications

This system can benefit multiple stakeholders:

- **Car Dealers**: Quick, consistent vehicle valuations
- **Buyers**: Informed negotiation with fair price estimates
- **Insurance Companies**: Accurate vehicle valuations for policies

- **Financial Institutions**: Loan collateral assessment
- **Online Marketplaces**: Automated price recommendations

IX. CONCLUSION AND FUTURE WORK

A. Conclusion

This research demonstrates the effectiveness of Random Forest ensemble learning for used car price prediction. Our model achieves an R^2 score of 0.9161 and MAE of €23,554.74, significantly outperforming the Linear Regression baseline. The comprehensive analysis of 120,000+ vehicle listings revealed that mileage, brand, and vehicle age are the primary pricing factors, collectively explaining over 95% of the model's predictive power.

The developed web application provides an accessible interface for real-time predictions, making the system practical for various automotive market applications. The combination of accurate predictions, interpretable feature importance, and user-friendly interface addresses real-world needs in vehicle valuation.

B. Future Research Directions

Several avenues exist for extending this work:

1) Advanced Algorithms:

- Gradient Boosting variants (XGBoost, LightGBM, CatBoost)
- Neural network architectures
- Hybrid ensemble methods

2) Enhanced Features:

- Vehicle condition reports and accident history
- Service and maintenance records
- Regional market trends and seasonal variations
- Optional equipment and specifications
- Vehicle images using computer vision

3) Time Series Integration:

- Temporal modeling of price trends
- Seasonal adjustment factors
- Market volatility prediction

4) Explainable AI:

- SHAP (SHapley Additive exPlanations) values for individual predictions
- LIME (Local Interpretable Model-agnostic Explanations)
- Counterfactual explanations for pricing scenarios

5) Multi-region Modeling:

- Region-specific models
- Transfer learning across markets
- Cultural and regulatory factor incorporation

6) Real-time Data Integration:

- Live market data feeds
- Automated model retraining pipelines
- Drift detection and adaptation

C. Final Remarks

The success of this machine learning approach to automotive price prediction demonstrates the value of data-driven methodologies in traditionally subjective domains. As data availability increases and algorithms improve, such systems will become increasingly integral to automotive commerce, benefiting all market participants through transparency and efficiency.

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